



Foundation in Information Technology (FIT)

FIT 103: ICT Applications
Module 1: Computing for the Society

Topic 5: Health Issues and Ergonomics



5. Health Issues and Ergonomics

This chapter explains the health issues that can arise due to incorrect and unhealthy practices when using computers and other digital devices. Then the chapter provides different approaches that can be used to minimise computer-related health issues.

5.1. Health Issues

The health issues described in this section provide an understanding of different physical problems people can experience when they use computers.

5.1.1. Musculoskeletal Problems

Have you ever experienced muscle pains when you were using the computer for a more extended period? This problem is known as Musculoskeletal Disorder (MSD). When you have a musculoskeletal disorder, you experience pains in your back, shoulder, wrist, and even you might get headaches. Repetitive stress to muscles, joints and nerves can cause musculoskeletal problems.

Why do you think we experience such health issues? It is mainly because of our negligence, and some of them are:

- Incorrect posture
- Poor support to the lower back
- Working long hours without stretching the body or taking breaks
- Poor ergonomics

There are two types of MSDs. They are Repetitive strain injuries (RSIs) and Back pain.

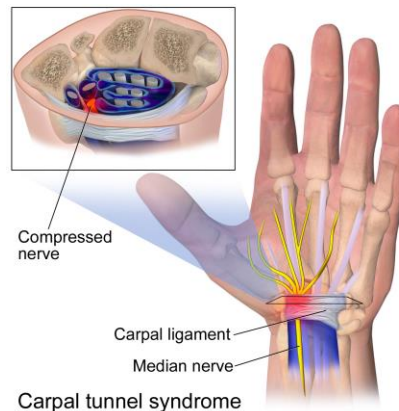
5.1.2. Repetitive strain injuries (RSIs)

- **A repetitive strain injury** is a damage that is progressed due to repeated movements of muscles and nerves. Mainly, arms are affected by RSI due to repetitive strain on the fingers, neck or shoulders of a person. The back pain develops with the incorrect postures. Following images illustrate images of incorrect postures that could lead to back pain.



5.1.3. Carpal Tunnel Syndrome (CTS)

Carpal Tunnel Syndrome is a common RSI condition. Carpal Tunnel Syndrome is experienced in the wrist; it causes pains, numbness and tingling. In the picture, you can see the median nerve. When this nerve is pressed unnecessarily, it causes strain in your hand. The main reason is that when you use the keyboard or mouse regularly, the median nerve gets a lot of pressure and starts to get compressed.



5.1.4. Computer Vision Syndrome (CVS)

Your eyes start to get strained when you use digital devices (computers, laptops, mobile phones, tablets, etc.) for a more extended period. As a result of that, you experience dry eyes, headaches, blur vision and sometimes pains in your neck and shoulder muscles. Poor lighting conditions, glare, poor ventilation and working without regular breaks cause CVS. The symptoms of CVS are:

- Dry eyes
- Eyestrain and Fatigue
- Blur vision
- Headache
- Pain in neck and shoulder

5.1.5. Tired and Stress

From the Oxygen we inhale, our brain uses 20% of it. When we stay at the computer for a long time, then the usage of the brain increases. Since there is no relaxation to the brain, it starts to get weary, which ultimately develops fatigue. Fatigue makes you feel exhausted, which causes stress. Stress affects both mental and physical health of humans.

5.2. Minimising Computer Related Health Issues

IT industry is a fast-growing industry around the world. Also, there are many users of digital devices. Therefore, it is crucial to understand how we can minimise health issues that

developed due to digital devices' usage. The sections from 5.2.1 to 5.2.6 explain how we can reduce the risks to our health.

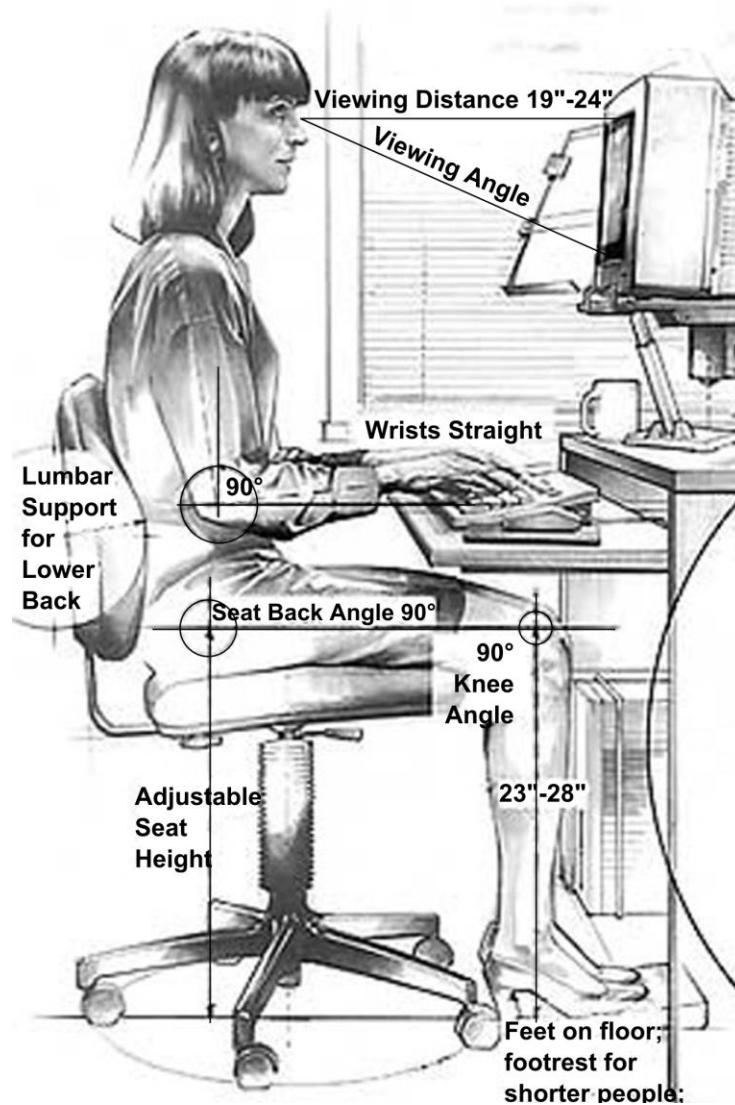
5.2.1. Ergonomics

Ergonomics is a word that evolved from the Greek dictionary. It means "laws of work". In North America, Ergonomics is referred to as "human factors". It is an approach that emphasises how people arrange or improve their workplaces and use the devices without causing risk or harm to themselves. Ergonomics is an important process in workplaces as it tries to promote the employees' health and safety.

5.2.2. Good Working Positions

Maintaining a good posture is important to mitigate most of the health issues mentioned in the previous sections. That is, good posture reduces strain on the back, arms, neck and shoulders. It would also be best if you remembered to change the posture as much as possible, allowing your body to stretch and take continuous breaks. Following are examples of good posture (see the picture below).

- When you sit, your body should be closer to the table.
- Head and neck should face forward without tilting to the sides of your body.
- Shoulders should be loosened (relaxed) and balanced.
- Hands, arms and wrists should be straight.
- Have support to your back and lower back.
- Feet should rest on the floor. If possible, use a footrest.



5.2.3. Lighting

Proper lighting can reduce eye strain and minimise headaches. Also, it is important to consider the level of lighting throughout the period you use the computer. When you use the computer, there should not be glare, dim or mirror images on your computer screen.

5.2.4. Glare

Glare is the reflection of surfaces when light directly hits on a computer screen. It causes eye strain and even headaches. To avoid glare, you can place your computer parallel to the window. Also, you can use blinds or curtains to control the natural light that comes into the room. You can use an anti-glare screen on your computer.

5.2.5. Ventilation

Air quality in a workplace is a significant factor for physical safety and wellbeing of employees. A wide variety of factors can cause weak indoor air quality. They are:

- Bacteria and microorganisms in the ventilation system
- Ozone and building insulation radiation
- Off-gassing and carpet pollution
- Cleaning substances
- Traffic Carbon Monoxide
- Smoking
- Over-occupied or poorly ventilated areas
- High humidity

The methods that can be used to improve workplace ventilation are:

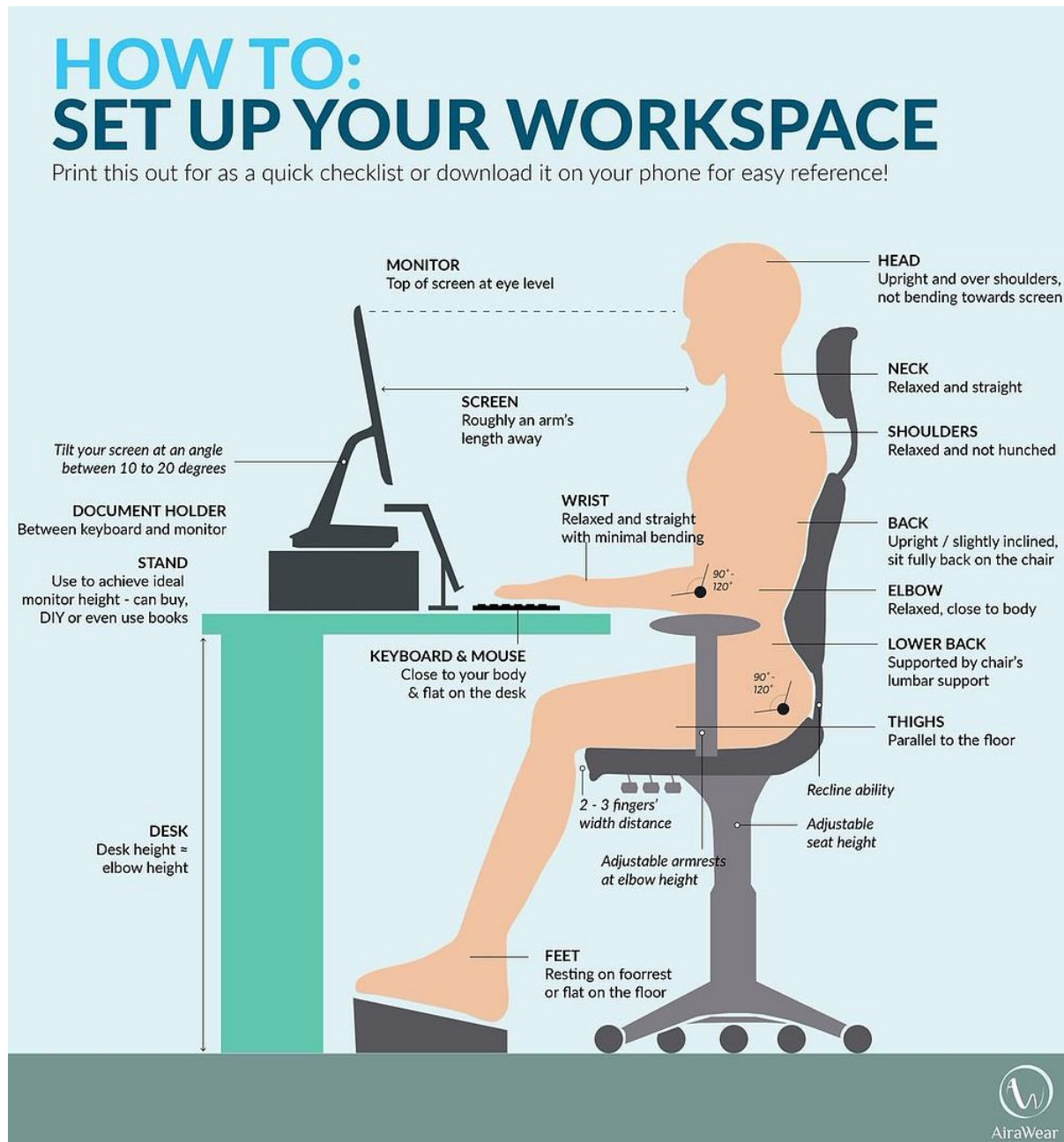
- Setting up oscillating fans to move air throughout the workplace
- Opening the windows
- Inspect heating and cooling systems regularly
- Install Carbon Monoxide detectors
- Clean the air ducts

5.2.6. Selecting and Arranging Workstation Components

In order to maintain a good posture, we can use the following steps to arrange the workstation (See the image).

- Adjusting the chair
 - When you adjust the seat, the feet need to be flat on the ground, and the knees should be placed on the same level or lower level than the hips.
 - The angle of the back of the chair should be 90 to 120 degrees.
 - Use the armrests to keep the shoulders relaxed.
 - Set the armrest to place the elbow and maintain 90 to 120 degrees.
- Adjusting the keyboard and mouse
 - Place the keyboard and mouse closer to your body.
 - Use a flat surface to place the keyboard and mouse.
- Adjust the computer screen
 - Top of the monitor should be at the eye level.
 - Place the screen an arm's length.
 - Tilt the monitor 10 to 20 degrees.

- Pauses and Breaks
 - Always take one to two minutes rest breaks every twenty minutes.
 - After an hour, try to change the task or stretch the body for five to ten minutes.
 - Change the eyesight by looking away from the monitor. During this, try to focus your eyes to an object in the distance.



Adapted from: <https://www.elevateergonomics.com/post/how-to-setup-an-ergonomic-workspace>

5.2.7. Ergonomics of using Mobile Devices

Keeping your phone close to your ear will cause issues with your wrists, neck and shoulder for long periods. You may also place undue stress on your fingers/thumbs, shoulders and neck by texting or using your mobile to create text dense emails and documents. Texting and using other mobile devices such as iPads can result in repetitive stress injury. The fingers and thumbs are not used to being used in this manner for a prolonged time.

Some precautions to be taken when using mobile devices are:

- Use hands-free devices to take phone calls.
- Do not only use your thumbs. Use fingers to type.
- Do not grip the phone too hard as it could strain your fingers and wrist.
- Maintain good posture by keeping the spine straight and do not bend the neck and the shoulders.